

Git & GitHub Cheat Sheet

Practical commands for college projects, coding practice, and GitHub submissions.

1. Git Setup

Command	Purpose
<code>git --version</code>	Check installed Git version.
<code>git config --global user.name "Lokesh"</code>	Set Git username.
<code>git config --global user.email "your@email.com"</code>	Set Git email.
<code>git config --list</code>	View Git configuration.

2. New Repository

Command	Purpose
<code>git init</code>	Convert current folder into a Git repository.
<code>git status</code>	Check current file/status information.

3. Add Files

Command	Purpose
<code>git add filename</code>	Add a specific file to staging.
<code>git add .</code>	Add all changed files to staging.
<code>git add *.cpp</code>	Add all C++ files.

4. Commit

Command	Purpose
<code>git commit -m "first commit"</code>	Save staged changes as a commit.
<code>git log</code>	View commit history.
<code>git log --oneline</code>	View compact commit history.

5. Connect to GitHub

Command	Purpose
<code>git remote -v</code>	Check connected remote repository.
<code>git remote add origin https://github.com/USERNAME/REPOSITORY.git</code>	Connect local repo to GitHub.
<code>git branch -M main</code>	Rename current branch to main.

6. Push to GitHub

Command	Purpose
<code>git push -u origin main</code>	First push; sets upstream tracking.
<code>git push</code>	Push later commits to GitHub.

7. Clone & Pull

Command	Purpose
<code>git clone https://github.com/USERNAME/REPOSITORY.git</code>	Download an existing GitHub repository.
<code>git pull</code>	Download and merge latest remote changes.
<code>git fetch</code>	Download remote information without merging.

8. Branches

Command	Purpose
<code>git branch</code>	List local branches.
<code>git branch feature</code>	Create a new branch.
<code>git switch feature</code>	Switch to a branch.
<code>git switch -c feature</code>	Create and immediately switch to a branch.
<code>git branch -d feature</code>	Delete a branch.

9. Merge

Command	Purpose
<code>git switch main</code>	Switch to main branch.
<code>git merge feature</code>	Merge feature branch into current branch.

10. Undo Changes

Command	Purpose
<code>git restore filename</code>	Discard unstaged changes in a file.
<code>git restore --staged filename</code>	Remove a file from staging.
<code>git reset --soft HEAD~1</code>	Undo last commit but keep changes staged.
<code>git reset --hard HEAD~1</code>	Undo last commit and discard changes; use carefully.

11. Useful Commands

Command	Purpose
<code>git diff</code>	See unstaged changes.
<code>git diff --staged</code>	See staged changes.
<code>git show</code>	View latest commit details.

Command	Purpose
<code>git branch -a</code>	View local and remote branches.
<code>git stash</code>	Temporarily save uncommitted changes.
<code>git revert COMMIT_ID</code>	Create a new commit that reverses an earlier commit.

■ Daily Git Workflow

```
git status
git add .
git commit -m "updated code"
git push
```

Git Flow

Your Code → git add . → Staging Area → git commit → Local Repository → git push → GitHub

GitHub → git pull → Local Repository → Your Files

■ Top 10 Commands to Remember

1. git init
2. git status
3. git add .
4. git commit -m "message"
5. git log
6. git clone
7. git remote -v
8. git push
9. git pull
10. git switch